Hridya Sharma

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Projects

HSNMatch – AI-Powered HSN/SAC Code Classifier
Tech: Python, Django REST Framework, PostgreSQL + pgvector, Jina AI, LLM API, React, Tailwind CSS
GitHub: github.com/SharmaHridya/HSNMATCHER | Live: hsnmatcher.netlify.app

- Built a **retrieval-first classifier** that maps plain-English product descriptions to HSN/SAC codes and GST rates, returning a confidence-ranked shortlist with reasoning.
- Embedded the HSN/SAC code list with Jina AI and indexed it in **PostgreSQL + pgvector**; vector search retrieves the **top-K** candidates deterministically before any LLM call.
- Limited the LLM to **re-ranking and explaining retrieved candidates**, so it cannot invent codes; a similarity threshold returns "unable to classify" for weak inputs and **near-ties are flagged as ambiguous**.
- Exposed a `POST /api/match/` REST endpoint; deployed the API on **Render** and the frontend on **Netlify**.

CartVerse – Full-Stack E-Commerce Platform
Tech: FastAPI, PostgreSQL, SQLAlchemy 2.x, Alembic, Pydantic v2, JWT, React (Vite), Tailwind CSS
GitHub: SharmaHridya/Ecommerce-frontend, Ecommerce-backend | Live: ecommerce-delta-rouge-32.vercel.app

- Built and deployed a full-stack store (**Vercel, Render, Neon**) with catalogue, category filtering, pagination, cart, checkout, order history and **verified-purchase reviews**.
- Implemented **JWT authentication with access and refresh tokens**, bcrypt password hashing and **role-based access control** for customer and admin routes.
- Made checkout atomic using `SELECT FOR UPDATE` **row locking** and **idempotent mock payments**; stock is validated server-side on every cart change to prevent overselling.
- Added an admin dashboard with inventory management and **low-stock alerts**, backed by a layered API (routes → schemas → CRUD → services) with Pydantic v2 and **Alembic migrations**.

Urban Heat Island Mitigation Simulator – ISRO Bharatiya Antariksh Hackathon 2026 (team of 4)
Tech: Python, Google Earth Engine, Landsat 8, Sentinel-2, Streamlit, Folium
GitHub: github.com/SharmaHridya/BAH-DATA

- Co-developed a geospatial **decision-support dashboard** that maps land surface temperature (LST) from **Landsat 8 Collection 2 Level 2** thermal imagery over multiple periods and tracks **5 urban heat hotspots** in Delhi NCR (Najafgarh, Bawana, Okhla, Palam, Badarpur).
- Computed **NDVI from Sentinel-2** imagery to quantify green cover, with switchable LST/NDVI layers and hotspot overlays in Streamlit and Folium.
- Developed an **intervention sandbox** estimating temperature reduction from tree-canopy expansion, cool roofs and surface-albedo enhancement using a simplified physics-inspired model.

Personal Expense Analyzer – Data Cleaning and Reporting Pipeline
Tech: Python, Pandas, Matplotlib
GitHub: github.com/SharmaHridya/expense_tracker_mini

- Wrote a Pandas pipeline that cleans messy expense CSVs: normalises category casing and payment modes, fills missing categories, **drops duplicate entries** and keeps refunds as negative amounts.
- Generates monthly, category-wise and payment-mode summaries, a category-by-month pivot table and the **top 5 expenses**, flags **budget overruns** per category per month, and saves a spending chart.

Technical Skills

Languages: Python, C, C++, JavaScript, SQL
Frontend: React.js, Vite, Tailwind CSS, HTML5, CSS3
Backend: FastAPI, Django REST Framework, Node.js, Express.js, REST APIs, JWT Authentication
Databases: PostgreSQL, pgvector, MongoDB, SQLAlchemy, Alembic
Data & AI: NumPy, Pandas, Matplotlib, Google Earth Engine, Streamlit, Folium, Vector Embeddings, LLM APIs, OpenCV (basic), scikit-learn (learning)
DevOps & Tools: Docker, Git, GitHub, Vercel, Render, Netlify, Jupyter

Education

Dr. B. R. Ambedkar National Institute of Technology, Jalandhar 2025 – 2029 (Expected)
B.Tech – Instrumentation and Control Engineering | **CGPA: 8.44**

Achievements

- Won an **Honourable Mention** and **Verbal Mentions** at inter-college Model United Nations conferences.
- Won **multiple inter-school and inter-college debate competitions**.
- Solved **230+ LeetCode problems** in data structures and algorithms.